Lesson 22: Fixed Point Theorems

AI/ML Mathematics

Contents

1 Motivation and Overview 2

2 References 2

3 The Banach Contraction Mapping Theorem 3

4 The Brouwer Fixed Point Theorem 4

5 The Schauder Fixed Point Theorem 5

6 The Kakutani Fixed Point Theorem 7

7 Hierarchy and Comparison 8

8 Exercises 9

9 Summary 10

10 Further Reading 11

1 Motivation and Overview

The equation $T(x) = x$—find a point that a map sends to itself—is one of the most universal formulations in mathematics. Differential equations, integral equations, game-theoretic equilibria, and the Bellman equation of dynamic programming can all be cast as fixed point problems. The theorems in this lesson provide conditions under which such equations have solutions, and in the most favorable case (the Banach contraction theorem) they also supply an algorithm and a convergence rate.

We develop four fixed point theorems of increasing generality:

1. The **Banach Contraction Mapping Theorem**, which guarantees existence, uniqueness, and geometric convergence of iterates for contractions on complete metric spaces.
2. The **Brouwer Fixed Point Theorem**, which guarantees existence (but not uniqueness) for continuous self-maps of compact convex subsets of $\mathbb{R}^n$.
3. The **Schauder Fixed Point Theorem**, which extends Brouwer to infinite-dimensional Banach spaces by requiring the map to be compact.
4. The **Kakutani Fixed Point Theorem**, which extends Brouwer to set-valued maps (correspondences), providing the standard route to Nash equilibrium existence.

Fixed Point Theorems in Reinforcement Learning

1. **Value iteration.** The Bellman optimality operator T is a γ-contraction on the space of bounded value functions. By the Banach contraction theorem, the iterates $V_{n+1} = TV_n$ converge geometrically to the unique optimal value function V^*, regardless of initialization.
2. **Policy iteration.** The policy evaluation step solves $V^\pi = T^\pi V^\pi$, which is again a contraction fixed point problem.
3. **Stationary distributions.** Brouwer's and Schauder's theorems guarantee the existence of stationary distributions for transition operators on compact state spaces, since the simplex of probability distributions is compact and convex.
4. **Nash equilibrium.** The best-response correspondence in a finite game satisfies the hypotheses of Kakutani's theorem on the product of mixed strategy simplices. The resulting fixed point is a Nash equilibrium —the foundational existence result of game theory and multi-agent RL.

Prerequisites. Metric spaces, completeness, and compactness (Lessons 15 and 18). Banach spaces and bounded linear operators (Lessons 16–17). The Baire Category Theorem (Lesson 18) is used in one remark but is not essential.

Outline. Section 2 lists references. Section 3 proves the Banach Contraction Mapping Theorem with the geometric convergence rate. Section 4 proves the Brouwer Fixed Point Theorem via the retraction argument. Section 5 extends Brouwer to infinite dimensions through the Schauder Fixed Point Theorem. Section 6 proves the Kakutani Fixed Point Theorem for set-valued maps. Section 7 summarizes the relationship between the four theorems. Section 8 provides exercises. Section 9 summarizes the key results. Section 10 offers further reading.

2 References

The Banach contraction mapping theorem is proved in Kreyszig [1989], Section 5.1, and Granas and Dugundji [2003], Chapter 1. The original paper is Banach [1922]. Brouwer's theorem appears in Granas and Dugundji [2003], Chapters 6–10, with the original formulation in Brouwer [1911]. The Schauder theorem is presented in Granas and Dugundji [2003], Chapter 8; the original is Schauder [1930]. Kakutani's theorem and its application to game theory are in Kakutani [1941]

and Granas and Dugundji [2003], Chapter 10. For the connection to Markov decision processes, see Puterman [2014], Chapter 6. General treatments of fixed point theory in functional analysis appear in Brézis [2011], Rudin [1991], and Dunford and Schwartz [1988].

3 The Banach Contraction Mapping Theorem

Definition 3.1 (Contraction Mapping)**.** Let (X, d) be a metric space. A map $T\colon X \to X$ is a *contraction* (or *contraction mapping*) if there exists a constant $\gamma \in [0, 1)$ such that

$$d(Tx, Ty) \leq \gamma\, d(x, y) \quad \text{for all } x, y \in X.$$

The constant γ is called the *contraction factor* (or *Lipschitz constant*). A contraction is automatically continuous (in fact, Lipschitz continuous).

Remark 3.2. A map satisfying $d(Tx, Ty) \leq d(x, y)$ (i.e., $\gamma = 1$) is called *nonexpansive*. Nonexpansive maps on complete metric spaces need not have fixed points (consider a translation on $\mathbb{R}$), and even when they do, iterates need not converge. The strict inequality $\gamma < 1$ is essential for the contraction theorem.

Theorem 3.3 (Banach Contraction Mapping Theorem)**.** *Let (X, d) be a nonempty complete metric space and $T\colon X \to X$ a contraction with factor $\gamma \in [0, 1)$. Then:*

(i) T has a unique fixed point $x^* \in X$ (i.e., $Tx^* = x^*$).

(ii) For any initial point $x_0 \in X$, the iterates $x_{n+1} = Tx_n$ converge to x^*.

(iii) The convergence is geometric: for all $n \geq 0$,

$$d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma}\, d(x_0, Tx_0).$$

Proof. **Existence and convergence.** Fix $x_0 \in X$ and define $x_{n+1} = Tx_n$ for $n \geq 0$. By the contraction property applied inductively,

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq \gamma\, d(x_n, x_{n-1}) \leq \cdots \leq \gamma^n\, d(x_1, x_0).$$

For $m > n \geq 0$, the triangle inequality gives

$$\begin{aligned} d(x_m, x_n) &\leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq \sum_{k=n}^{m-1} \gamma^k\, d(x_1, x_0) \\ &\leq \gamma^n \sum_{j=0}^{\infty} \gamma^j\, d(x_1, x_0) = \frac{\gamma^n}{1-\gamma}\, d(x_1, x_0). \end{aligned} \tag{1}$$

Since $\gamma < 1$, the right-hand side tends to 0 as $n \to \infty$, so (x_n) is a Cauchy sequence. By completeness of X, there exists $x^* \in X$ with $x_n \to x^*$.

Since T is Lipschitz (hence continuous),

$$Tx^* = T\Big(\lim_{n\to\infty} x_n\Big) = \lim_{n\to\infty} Tx_n = \lim_{n\to\infty} x_{n+1} = x^*.$$

Uniqueness. Suppose $Tx^* = x^*$ and $Ty^* = y^*$. Then $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \gamma\, d(x^*, y^*)$. Since $\gamma < 1$, this forces $(1-\gamma)\, d(x^*, y^*) \leq 0$, so $d(x^*, y^*) = 0$ and $x^* = y^*$.

Geometric convergence rate. Let $m \to \infty$ in (1):

$$d(x_n, x^*) = \lim_{m\to\infty} d(x_m, x_n) \leq \frac{\gamma^n}{1-\gamma}\, d(x_1, x_0) = \frac{\gamma^n}{1-\gamma}\, d(x_0, Tx_0). \qquad \square$$

Banach Contraction Mapping Theorem

In a complete metric space, every contraction T with factor $\gamma < 1$ has a *unique* fixed point x^*. Iterating T from any starting point converges to x^* at the geometric rate $d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma} d(x_0, Tx_0)$. After $n \geq \frac{1}{1-\gamma} \log \frac{d(x_0,Tx_0)}{(1-\gamma)\varepsilon}$ iterations, $d(x_n, x^*) \leq \varepsilon$.

Value Iteration as a Contraction

Let $\mathcal{S}$ be a finite state space and consider the Banach space $(B(\mathcal{S}), \|\cdot\|_\infty)$ of bounded real-valued functions on $\mathcal{S}$. The **Bellman optimality operator**

$$(TV)(s) = \max_{a \in \mathcal{A}} \Big[R(s,a) + \gamma \sum_{s'} P(s' \mid s, a)\, V(s') \Big]$$

is a γ-contraction in $\|\cdot\|_\infty$ (this is proved in Exercise 8.1). By the Banach contraction theorem:

- There exists a unique V^* with $TV^* = V^*$ (the Bellman equation).
- Value iteration $V_{n+1} = TV_n$ converges to V^* from any initialization.
- The iteration count for ε-optimality is $O\Big(\frac{1}{1-\gamma} \log \frac{1}{(1-\gamma)\varepsilon}\Big)$.

Policy evaluation (T^π instead of T) is likewise a contraction, so the same guarantees apply.

Example 3.4 (Picard–Lindelof via Contraction)**.** Consider the initial value problem $y'(t) = f(t, y(t))$, $y(0) = y_0$, where f is Lipschitz in y with constant L. The equivalent integral equation $y(t) = y_0 + \int_0^t f(s, y(s))\, ds$ defines an operator T on $C([0,h])$ for $h > 0$. On a sufficiently short interval ($h < 1/L$), the operator T is a contraction in $\|\cdot\|_\infty$, giving existence and uniqueness of the solution. More precisely, by using the weighted norm $\|y\|_w = \sup_{t\in[0,h]} e^{-2Lt}|y(t)|$, one can show T is a contraction with factor $1/2$ on any interval, thereby recovering the global Picard–Lindelof theorem.

4 The Brouwer Fixed Point Theorem

The Banach contraction theorem requires a quantitative condition ($\gamma < 1$) that is too strong for many applications. The Brouwer theorem drops the contraction requirement entirely, asking only that the map be continuous and the domain be compact and convex. The price is that we lose uniqueness and any constructive iteration scheme.

Theorem 4.1 (Brouwer Fixed Point Theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be a nonempty compact convex set and $f\colon C \to C$ a continuous map. Then f has a fixed point: there exists $x^* \in C$ with $f(x^*) = x^*$.*

We prove the theorem for $C = \overline{B}^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ (the closed unit ball). The general case follows because every nonempty compact convex subset of $\mathbb{R}^n$ is homeomorphic to a closed ball of the appropriate dimension.

The proof proceeds by contradiction, using the no-retraction lemma.

Definition 4.2 (Retraction)**.** Let $A \subseteq X$ be a subspace of a topological space X. A *retraction* of X onto A is a continuous map $r\colon X \to A$ such that $r(a) = a$ for all $a \in A$.

Lemma 4.3 (No-Retraction Lemma)**.** *There is no retraction of the closed ball $\overline{B}^n$ onto the sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$.*

Proof. We give the proof for $n = 2$ via a smooth approximation and degree argument; the general case is analogous.

Suppose $r\colon \overline{B}^2 \to S^1$ is a retraction. By the Weierstrass approximation theorem, r can be uniformly approximated by smooth maps. Composing with the normalization map $x \mapsto x/\|x\|$ (which is smooth away from the origin), one obtains a smooth map $\tilde{r}\colon \overline{B}^2 \to S^1$ with $\tilde{r}|_{S^1}$ homotopic to the identity on S^1.

The *degree* of the identity map $\mathrm{id}\colon S^1 \to S^1$ is 1. However, any smooth map $\overline{B}^2 \to S^1$ that extends to the whole disk must have degree 0 on the boundary (since the disk is contractible, any map from it to S^1 can be lifted to a map into $\mathbb{R}$, and such lifted maps have winding number 0).

Since $\tilde{r}|_{S^1}$ is homotopic to id_{S^1} (which has degree 1) but must have degree 0, we reach a contradiction. Hence no such retraction exists. □

Proof of the Brouwer Fixed Point Theorem (Theorem 4.1). Suppose for contradiction that $f(x) \neq x$ for all $x \in \overline{B}^n$. For each $x \in \overline{B}^n$, consider the ray starting at $f(x)$ and passing through x. Since $f(x) \neq x$, this ray is well-defined. Let $r(x)$ be the unique point where this ray intersects the sphere S^{n-1} on the side of x (i.e., $r(x) = x + t(x - f(x))$ for the unique $t \geq 0$ with $\|r(x)\| = 1$).

Continuity of r. The parameter $t = t(x)$ is determined by the equation $\|x+t(x-f(x))\|^2 = 1$, which expands to a quadratic in t with coefficients depending continuously on x (since f is continuous). The discriminant is strictly positive (because $f(x) \neq x$ guarantees the ray from $f(x)$ through x exits the ball), so $t(x)$ is a continuous function of x by the continuous dependence of roots of polynomials on their coefficients. Hence r is continuous.

r is a retraction. For $x \in S^{n-1}$, we have $\|x\| = 1$, so $t = 0$ satisfies $\|x + 0 \cdot (x - f(x))\| = \|x\| = 1$. Thus $r(x) = x$ for all $x \in S^{n-1}$.

Therefore $r\colon \overline{B}^n \to S^{n-1}$ is a retraction, contradicting Lemma 4.3. The assumption that f has no fixed point must be false. □

Remark 4.4. The Brouwer theorem is a pure existence result and provides no algorithm for finding the fixed point. Computing approximate Brouwer fixed points is PPAD-complete—widely believed to be intractable. When the contraction structure of the Banach theorem is available, it provides the far more useful algorithmic guarantee.

Stationary Distributions via Brouwer

Let P be the transition matrix of a Markov chain on a finite state space $\mathcal{S}$ with $|\mathcal{S}| = n$. A stationary distribution is a vector $\mu \in \Delta_n$ (the probability simplex) satisfying $\mu^\top P = \mu^\top$. Equivalently, μ is a fixed point of the map $f(\mu) = \mu^\top P$ restricted to Δ_n. Since Δ_n is a nonempty compact convex subset of $\mathbb{R}^n$ and f is linear (hence continuous) with $f(\Delta_n) \subseteq \Delta_n$ (the image of a probability vector under a stochastic matrix is again a probability vector), the Brouwer theorem guarantees the existence of at least one stationary distribution.

5 The Schauder Fixed Point Theorem

In infinite-dimensional Banach spaces, the closed unit ball is not compact (by Riesz's lemma), so the Brouwer theorem does not directly apply. The Schauder theorem overcomes this obstacle by requiring the map to be *compact*—a condition that recovers enough finite-dimensional structure to invoke Brouwer.

Definition 5.1 (Compact Operator)**.** Let X be a Banach space and $C \subseteq X$. A map $T\colon C \to X$ is *compact* if T is continuous and maps bounded subsets of C to relatively compact subsets of X (i.e., subsets whose closure is compact).

Remark 5.2. Every continuous map with finite-dimensional range is compact (since bounded subsets of finite-dimensional spaces are relatively compact). Compact operators are the "next

best thing" to finite-dimensional operators, and they arise naturally as integral operators with continuous kernels (by the Arzela–Ascoli theorem).

Theorem 5.3 (Schauder Fixed Point Theorem)**.** *Let X be a Banach space, $C \subseteq X$ a nonempty closed bounded convex set, and $T\colon C \to C$ a compact map. Then T has a fixed point.*

Proof. Since T is compact and C is bounded, the set $K = \overline{T(C)}$ is compact. We approximate T by finite-dimensional maps and apply the Brouwer theorem.

Step 1: Schauder projections. For each $n \in \mathbb{N}$, cover K by finitely many open balls of radius $1/n$: there exist points $y_1^{(n)}, \dots, y_{k_n}^{(n)} \in T(C)$ such that $K \subseteq \bigcup_{i=1}^{k_n} B(y_i^{(n)}, 1/n)$. Define the weight functions

$$\lambda_i^{(n)}(x) = \max\bigl(0,\ 1/n - \|x - y_i^{(n)}\|\bigr), \qquad i = 1, \dots, k_n.$$

For $x \in K$, at least one ball covers x, so $\sum_{i=1}^{k_n} \lambda_i^{(n)}(x) > 0$. The *Schauder projection* $P_n\colon K \to X$ is

$$P_n(x) = \frac{\sum_{i=1}^{k_n} \lambda_i^{(n)}(x)\, y_i^{(n)}}{\sum_{i=1}^{k_n} \lambda_i^{(n)}(x)}.$$

Then P_n is continuous, its range lies in $C_n = \operatorname{conv}\{y_1^{(n)}, \dots, y_{k_n}^{(n)}\} \subseteq C$ (since C is convex and $y_i^{(n)} \in T(C) \subseteq C$), and

$$\|P_n(x) - x\| < 1/n \quad \text{for all } x \in K. \tag{2}$$

Indeed, $\lambda_i^{(n)}(x) > 0$ only when $\|x - y_i^{(n)}\| < 1/n$, so $P_n(x)$ is a convex combination of points within distance $1/n$ of x.

Step 2: Finite-dimensional fixed points. The set C_n is a compact convex subset of a finite-dimensional subspace of X. Define $T_n\colon C_n \to C_n$ by $T_n = P_n \circ T|_{C_n}$. This is a continuous self-map of a compact convex set in finite dimensions. By the Brouwer Fixed Point Theorem 4.1, there exists $x_n \in C_n$ with $T_n(x_n) = x_n$, i.e.,

$$x_n = P_n(Tx_n).$$

By (2), $\|x_n - Tx_n\| = \|P_n(Tx_n) - Tx_n\| < 1/n$.

Step 3: Passing to the limit. The sequence $(x_n)_{n\geq 1}$ lies in C (which is bounded), so $(Tx_n)_{n\geq 1}$ lies in $T(C)$, whose closure K is compact. Hence (Tx_n) has a convergent subsequence, say $Tx_{n_j} \to y \in K$. Since $\|x_{n_j} - Tx_{n_j}\| < 1/n_j \to 0$, we also have $x_{n_j} \to y$. Since C is closed, $y \in C$. By continuity of T,

$$Ty = T\left(\lim_{j\to\infty} x_{n_j}\right) = \lim_{j\to\infty} Tx_{n_j} = y. \qquad \square$$

Remark 5.4. A more general version states: if C is a nonempty convex subset of a Banach space (not necessarily closed or bounded) and $T\colon C \to C$ is continuous with $T(C)$ relatively compact, then T has a fixed point. The proof is similar—the closure of $T(C)$ is compact, and the argument of Steps 1–3 applies with minor modifications.

Schauder and Continuous MDPs

In continuous-state MDPs, the state space $\mathcal{S}$ is a compact subset of $\mathbb{R}^d$ and value functions live in the Banach space $C(\mathcal{S})$. When the transition kernel has a continuous density, the Bellman evaluation operator T^π maps bounded sets of continuous functions to equicontinuous families (by the integral smoothing effect), so T^π is compact by the Arzela–Ascoli theorem. The Schauder theorem then guarantees the existence of a fixed point $V^\pi \in C(\mathcal{S})$—the value function of policy π. When the discount factor $\gamma < 1$, the

Banach contraction theorem also applies and gives the stronger conclusion of uniqueness and geometric convergence; the Schauder theorem is relevant in undiscounted settings where the contraction property fails.

6 The Kakutani Fixed Point Theorem

The Brouwer and Schauder theorems apply to single-valued maps. In game theory and mathematical economics, one frequently encounters *set-valued maps* (correspondences), such as the best-response correspondence that assigns to each opponent strategy profile the set of optimal strategies. Kakutani's theorem extends Brouwer's result to this setting.

Definition 6.1 (Set-Valued Map / Correspondence)**.** A *set-valued map* (or *correspondence*) from a set X to a set Y is a map $\Phi\colon X \rightrightarrows Y$ that assigns to each $x \in X$ a subset $\Phi(x) \subseteq Y$. A *fixed point* of $\Phi\colon X \rightrightarrows X$ is a point x^* with $x^* \in \Phi(x^*)$.

Definition 6.2 (Upper Hemicontinuity)**.** A set-valued map $\Phi\colon X \rightrightarrows Y$ between topological spaces is *upper hemicontinuous* (u.h.c.) if for every open set $V \subseteq Y$, the upper inverse $\Phi^+(V) = \{x \in X : \Phi(x) \subseteq V\}$ is open in X.

Equivalently, when X and Y are metric spaces and Φ has compact values: Φ is u.h.c. if and only if it has a closed graph, meaning that $x_n \to x$, $y_n \in \Phi(x_n)$, and $y_n \to y$ together imply $y \in \Phi(x)$.

Remark 6.3. A correspondence is *lower hemicontinuous* if for every open $V \subseteq Y$, the lower inverse $\Phi^-(V) = \{x \in X : \Phi(x) \cap V \neq \emptyset\}$ is open. Kakutani's theorem requires upper hemicontinuity; lower hemicontinuity is not needed.

Theorem 6.4 (Kakutani Fixed Point Theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be a nonempty compact convex set and $\Phi\colon C \rightrightarrows C$ a set-valued map satisfying:*

(i) For every $x \in C$, the set $\Phi(x)$ is nonempty and convex.

(ii) Φ is upper hemicontinuous.

Then Φ has a fixed point: there exists $x^* \in C$ with $x^* \in \Phi(x^*)$.

Proof. We reduce to the Brouwer theorem by approximating Φ with continuous single-valued maps.

Step 1: Approximate selections. For each $k \in \mathbb{N}$, since C is compact, we can cover it by finitely many open balls $\{B(x_i, 1/k)\}_{i=1}^{m_k}$ with centers $x_i \in C$. Let $\{\lambda_i\}_{i=1}^{m_k}$ be a continuous partition of unity subordinate to this cover, i.e., each $\lambda_i\colon C \to [0,1]$ is continuous, $\operatorname{supp}(\lambda_i) \subseteq B(x_i, 1/k)$, and $\sum_{i=1}^{m_k} \lambda_i(x) = 1$ for all $x \in C$.

For each i, choose any $y_i \in \Phi(x_i)$ (possible since $\Phi(x_i)$ is nonempty). Define the continuous function

$$f_k(x) = \sum_{i=1}^{m_k} \lambda_i(x)\, y_i.$$

Since each $y_i \in C$ and C is convex, $f_k(x)$ is a convex combination of points in C, so $f_k\colon C \to C$.

Step 2: Apply the Brouwer theorem. By Theorem 4.1, each f_k has a fixed point $x_k \in C$ with $f_k(x_k) = x_k$.

Step 3: Pass to the limit. Since C is compact, (x_k) has a convergent subsequence. Passing to this subsequence (still denoted (x_k)), let $x_k \to x^*$. We show $x^* \in \Phi(x^*)$.

Let V be any open convex set with $\Phi(x^*) \subseteq V$. By upper hemicontinuity, $\Phi^+(V) = \{x \in C : \Phi(x) \subseteq V\}$ is open and contains x^*. Since $x_k \to x^*$, for all sufficiently large k:

- $x_k \in \Phi^+(V)$, so $\Phi(x_k) \subseteq V$.
- $\|x_k - x^*\| < 1/k$.

For such large k, if $\lambda_i(x_k) > 0$ then $\|x_k - x_i\| < 1/k$, hence $\|x_i - x^*\| < 2/k$. For k large enough, x_i lies in $\Phi^+(V)$ as well, so $\Phi(x_i) \subseteq V$, and in particular $y_i \in V$.

Therefore $x_k = f_k(x_k) = \sum_i \lambda_i(x_k)\, y_i$ is a convex combination of points $y_i \in V$. Since V is convex, $x_k \in V$.

Since this holds for every open convex $V \supseteq \Phi(x^*)$ and $x_k \to x^*$, we conclude $x^* \in \overline{V}$ for every such V. Because $\Phi(x^*)$ is compact and convex,

$$x^* \in \bigcap \left\{ \overline{V} : V \supseteq \Phi(x^*),\ V \text{ open and convex} \right\} = \Phi(x^*). \qquad \square$$

Nash Equilibrium Existence via Kakutani

Consider a finite n-player game. Player i has a finite action set A_i and a payoff function $u_i \colon \prod_{j=1}^n A_j \to \mathbb{R}$. A *mixed strategy* for player i is $\sigma_i \in \Delta(A_i)$ (the simplex of probability distributions over A_i). The *best-response correspondence* of player i is

$$BR_i(\sigma_{-i}) = \arg\max_{\sigma_i \in \Delta(A_i)} \sum_{a \in A} \left(\prod_{j=1}^n \sigma_j(a_j) \right) u_i(a),$$

where $\sigma_{-i} = (\sigma_j)_{j \neq i}$. Define the joint best-response $\Phi(\sigma) = \prod_{i=1}^n BR_i(\sigma_{-i})$ on the compact convex set $C = \prod_{i=1}^n \Delta(A_i)$.

One verifies:

1. Each $BR_i(\sigma_{-i})$ is nonempty and convex (it is the set of maximizers of a linear function over a simplex).
2. Φ is upper hemicontinuous (by the Berge Maximum Theorem: the objective is continuous and the constraint set is constant).

By Kakutani's theorem, Φ has a fixed point σ^*, which is precisely a **Nash equilibrium**: every player is best-responding to the others. This is the original proof of Kakutani [1941] as applied by Nash (1950).

7 Hierarchy and Comparison

The four fixed point theorems form a natural hierarchy, summarized in the following table.

Theorem	**Setting**	**Map requirement**	**Conclusion**
Banach	Complete metric space	Contraction ($\gamma < 1$)	Unique FP + algorithm
Brouwer	Compact convex $\subseteq \mathbb{R}^n$	Continuous, single-valued	Existence
Schauder	Closed bdd. convex in Banach sp.	Compact (continuous + bdd $\to$ rel. cpt.)	Existence
Kakutani	Compact convex $\subseteq \mathbb{R}^n$	u.h.c., convex-valued corresp.	Existence

Remark 7.1. The logical dependencies are:

$$\text{Brouwer} \xrightarrow{\text{compact approx.}} \text{Schauder} \qquad \text{and} \qquad \text{Brouwer} \xrightarrow{\text{approx. selections}} \text{Kakutani.}$$

The Banach theorem is independent of the others: it uses completeness and a metric contraction condition, while the topological theorems use compactness and convexity. The Banach theorem gives uniqueness and an algorithm; the others give only existence.

Choosing the Right Theorem

- Use **Banach** when you have a complete metric space and a contraction—this is the case for value iteration with $\gamma < 1$.
- Use **Brouwer** for existence arguments in finite-dimensional compact convex sets (e.g., stationary distributions, equilibria in simplex-valued problems).
- Use **Schauder** when working in infinite-dimensional spaces with compact operators (e.g., integral operators on function spaces).
- Use **Kakutani** when the "solution map" is set-valued (e.g., best-response correspondences in game theory).

8 Exercises

Exercise 8.1 (Bellman Operator Is a Contraction (**))**.** Let $\mathcal{S}$ and $\mathcal{A}$ be finite sets, $R\colon \mathcal{S}\times\mathcal{A} \to \mathbb{R}$ a reward function, $P(\cdot \mid s, a)$ a transition kernel, and $\gamma \in (0, 1)$ a discount factor. Define the Bellman optimality operator T on the space $B(\mathcal{S})$ of bounded functions on $\mathcal{S}$ (equipped with $\|\cdot\|_\infty$) by

$$(TV)(s) = \max_{a\in\mathcal{A}} \Big[R(s, a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s, a)\, V(s')\Big].$$

(a) Prove that T is a γ-contraction: $\|TV - TW\|_\infty \le \gamma\, \|V - W\|_\infty$.

Hint: For each s, let a_s^* achieve the maximum for $(TV)(s)$. Then $(TV)(s) - (TW)(s) \le \gamma \sum_{s'} P(s' \mid s, a_s^*)(V(s') - W(s'))$. By symmetry, obtain the reverse inequality and take the supremum over s.

(b) Deduce from the Banach contraction theorem that value iteration converges geometrically and that V^* is the unique solution of the Bellman equation $TV = V$.

(c) How many iterations n are needed to guarantee $\|V_n - V^*\|_\infty \le \varepsilon$? Express the answer in terms of γ, ε, and $\|V_1 - V_0\|_\infty$.

Exercise 8.2 (Contractive Iterate (**))**.** Let (X, d) be a complete metric space and $T\colon X \to X$ a continuous map such that T^N is a contraction for some $N \ge 1$.

(a) Show that T^N has a unique fixed point x^*.

(b) Show that Tx^* is also a fixed point of T^N, and conclude $Tx^* = x^*$.

(c) Deduce that T itself has a unique fixed point.

Application: The Picard operator for an ODE on $[0, h]$ may not be a contraction in $\|\cdot\|_\infty$, but T^N is for N large enough. This exercise shows uniqueness still holds.

Exercise 8.3 (Brouwer in One Dimension (*))**.** (a) Prove the Brouwer Fixed Point Theorem for $n = 1$ directly: if $f\colon [a, b] \to [a, b]$ is continuous, then f has a fixed point.

Hint: Apply the intermediate value theorem to $g(x) = f(x) - x$.

(b) Show by example that the conclusion fails if $[a, b]$ is replaced by an open interval (a, b).

(c) Show by example that the conclusion fails if $[a, b]$ is replaced by a non-convex compact set $K = [0, 1] \cup [2, 3]$.

Exercise 8.4 (Schauder and Integral Equations (**))**.** Let $K\colon [0, 1] \times [0, 1] \to \mathbb{R}$ be continuous and define $T\colon C[0, 1] \to C[0, 1]$ by $(Tf)(t) = \int_0^1 K(t, s)\, f(s)\, ds$.

(a) Show that T maps the closed ball $\overline{B}(0,R) = \{f \in C[0,1] : \|f\|_\infty \le R\}$ into itself for R sufficiently large, provided we add a continuous inhomogeneity: $(Tf)(t) = g(t) + \int_0^1 K(t,s)\, f(s)\, ds$. Find the condition on R in terms of $\|g\|_\infty$ and $\|K\|_\infty$.

(b) Show that $T(\overline{B}(0,R))$ is equicontinuous.

Hint: Use uniform continuity of K on $[0,1]^2$.

(c) Conclude via the Arzela–Ascoli theorem that T is a compact operator.

(d) Apply the Schauder theorem to deduce that the integral equation $f = Tf$ has a solution.

Exercise 8.5 (Kakutani and Two-Player Zero-Sum Games (**))**.** Let $A \in \mathbb{R}^{m\times n}$ be a payoff matrix for a two-player zero-sum game. Player 1 chooses $p \in \Delta_m$ and Player 2 chooses $q \in \Delta_n$, with payoff $p^\top Aq$ to Player 1.

(a) Write the best-response correspondences $BR_1(q)$ and $BR_2(p)$ explicitly.

(b) Verify the hypotheses of the Kakutani theorem for the joint best-response $\Phi(p,q) = BR_1(q) \times BR_2(p)$ on $\Delta_m \times \Delta_n$.

(c) Conclude that a Nash equilibrium (saddle point) exists.

(d) Explain how this relates to the minimax theorem $\max_p \min_q p^\top Aq = \min_q \max_p p^\top Aq$.

Exercise 8.6 (No Fixed Point Without Compactness (*))**.** Give an example of a continuous map $f\colon C \to C$ on a closed convex but *unbounded* subset C of $\mathbb{R}^n$ that has no fixed point. This shows that compactness in the Brouwer theorem cannot be dropped.

Hint: Consider a translation on $C = \mathbb{R}$.

Exercise 8.7 (Policy Iteration and Contraction (***))**.** In the MDP setting of Exercise 8.1, *policy iteration* generates a sequence of policies $\pi_0, \pi_1, \ldots$ by: (i) evaluate V^{π_k} as the unique fixed point of T^{π_k} (where $(T^\pi V)(s) = R(s,\pi(s)) + \gamma \sum_{s'} P(s' \mid s,\pi(s))\, V(s')$), and (ii) improve: $\pi_{k+1}(s) = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^{\pi_k}(s')]$.

(a) Show that T^π is a γ-contraction for any policy π, so V^π is well-defined and unique.

(b) Show that $V^{\pi_{k+1}} \ge V^{\pi_k}$ componentwise.

Hint: $T^{\pi_{k+1}} V^{\pi_k} \ge T^{\pi_k} V^{\pi_k} = V^{\pi_k}$. Now iterate.

(c) Conclude that policy iteration converges to V^* in finitely many steps (there are finitely many deterministic policies).

9 Summary

- **Contraction.** A map T on a metric space with $d(Tx,Ty) \le \gamma\, d(x,y)$ for some $\gamma < 1$.
- **Banach Contraction Mapping Theorem.** On a complete metric space, every contraction has a unique fixed point. The iterates $x_{n+1} = Tx_n$ converge from any starting point at geometric rate $d(x_n, x^*) \le \frac{\gamma^n}{1-\gamma}\, d(x_0, Tx_0)$. This is the theoretical foundation of value iteration.
- **Brouwer Fixed Point Theorem.** Every continuous self-map of a nonempty compact convex subset of $\mathbb{R}^n$ has a fixed point. Proved via the no-retraction lemma.
- **Compact operator.** A continuous map that sends bounded sets to relatively compact sets.
- **Schauder Fixed Point Theorem.** Every compact self-map of a nonempty closed bounded convex subset of a Banach space has a fixed point. Proved by approximating with finite-dimensional maps (Schauder projections) and applying Brouwer.

- **Set-valued map (correspondence).** $\Phi\colon X \rightrightarrows Y$ assigns a subset $\Phi(x) \subseteq Y$ to each x. Upper hemicontinuity requires $\{x : \Phi(x) \subseteq V\}$ to be open for every open V.
- **Kakutani Fixed Point Theorem.** An upper hemicontinuous correspondence with nonempty convex values on a compact convex subset of $\mathbb{R}^n$ has a fixed point. This is the theorem used to prove Nash equilibrium existence.
- **Hierarchy.** Banach gives uniqueness and an algorithm; Brouwer/Schauder/Kakutani give only existence. Brouwer implies Schauder (via compact approximation) and Kakutani (via approximate selections).

10 Further Reading

The definitive reference on fixed point theory is Granas and Dugundji [2003], which covers all four theorems with detailed proofs, historical notes, and numerous applications. The Banach contraction theorem in the context of metric spaces is also treated in Kreyszig [1989], Section 5.1, and Rudin [1991]. For a proof of the Brouwer theorem via Sperner's lemma (a combinatorial alternative to the retraction argument), see Granas and Dugundji [2003], Chapter 6.

The connection between the Banach contraction theorem and Markov decision processes is developed rigorously in Puterman [2014], Chapter 6. The application of Kakutani's theorem to Nash equilibrium existence follows Kakutani [1941].

The preceding lesson (Lesson 21) proved the four fundamental theorems of functional analysis (Hahn–Banach, Uniform Boundedness, Open Mapping, Closed Graph). The next topic in the curriculum (Lesson 23) begins the optimization sequence with convex sets and their structural properties.

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